Skills Summary

Modeling Loci and Intersections with Conic Sections

Use this reference while completing your worksheet or when studying.

Two moves cover this whole unit. A *locus* is a sentence about distance: write the distances, set them equal (or in the stated ratio), square, simplify. An *intersection* is a system: substitute, solve one variable, substitute back for the other.

1. Circles from a distance condition

Center-radius form: $(x-h)^2+(y-k)^2=r^2$ has center (h,k) and radius r. The signs in the brackets are the *opposite* of the center coordinates: a center of $(3,-4)$ gives $(x-3)^2+(y+4)^2$.

From general form, complete the square in x and in y: halve each linear coefficient, square it, and add it to *both* sides. The radius is the square root of the new right-hand side, never of the original constant.

Example: Write $x^2+y^2+10x-4y+13=0$ in center-radius form.
Step 1. The equation has no visible center, so completing the square in each variable is the tool that exposes one.
Step 2. Move 13 across and add $(10/2)^2=25$ and $(-4/2)^2=4$ to both sides: $(x+5)^2+(y-2)^2=-13+25+4=16$, so $r=\sqrt{16}$.
center $\boldsymbol{(-5,\ 2)}$, $\boldsymbol{r=4}$

Try it: Write $x^2+y^2-2x+6y+6=0$ in center-radius form.

check: center $\boldsymbol{(1,\ -3)}$, $\boldsymbol{r=2}$

2. Parabolas from a focus and a directrix

Definition: a parabola is every point whose distance to the *focus* equals its distance to the *directrix*.

With vertex at the origin, focus $(0,p)$ and directrix $y=-p$, that condition simplifies to $x^2=4py$, so $y=x^2/(4p)$.

(!) p is the focal *distance*; the number in the equation is $4p$. A focus at $(0,3)$ gives $x^2=12y$, not $x^2=3y$.

Example: A parabola has focus $(0,4)$ and directrix $y=-4$. Find y when $x=8$.
Step 1. The focus is 4 above the vertex and the directrix 4 below it, so $p=4$ and the standard form $x^2=4py$ applies directly.
Step 2. $x^2=16y$, so $y=8^2/16=64/16$.
4

Try it: A parabola has focus $(0,2)$ and directrix $y=-2$. Find y when $x=8$.

check: **8**

3. Where a line meets a conic

Substitute, do not guess. A line solved for y drops straight into the conic and leaves one quadratic in x.

The root count is geometry: two real roots means the line cuts the curve twice (a secant); one repeated root means it touches once (a tangent); no real roots means it misses. Check the discriminant if the question asks about tangency.

Finish the job: put each root back into the *line* — the easier equation — and report ordered pairs.

Example: Where does $y = x - 1$ meet $x^2 + y^2 = 13$?
Step 1. The line is already solved for y, so substituting it into the circle turns a two-variable system into one quadratic.
Step 2. $x^2 + (x-1)^2 = 13$ gives $2x^2 - 2x - 12 = 0$, so $x^2 - x - 6 = 0$ and $(x-3)(x+2) = 0$; then $y = x - 1$ gives 2 and -3.
$(3,\ 2)$ and **$(-2,\ -3)$**

Try it: Where does $y = x + 2$ meet $x^2 + y^2 = 10$?

check: **$(1,\ 3)$** and **$(-3,\ -1)$**

4. Where two conics meet

Eliminate the squared variable, not the linear one. If one curve gives x^2 in terms of y (a parabola does), substitute that whole expression into the other equation — you get a quadratic in y with no square roots anywhere.

For $xy = c$, substitute $y = c/x$ and clear the denominator; the result is a quadratic in x^2.

(!) Test every value you find. A y that forces $x^2 < 0$ gives no real point, and $x^2 = 0$ gives only one. Two conics can meet in 0, 1, 2, 3 or 4 points.

Example: Where does $x^2 + y^2 = 25$ meet $y = x^2 - 13$?
Step 1. The parabola gives $x^2 = y + 13$, so substituting for x^2 removes x completely and leaves a quadratic in y.
Step 2. $(y + 13) + y^2 = 25$ gives $y^2 + y - 12 = 0$, so $y = 3$ or $y = -4$; then $x^2 = 16$ and $x^2 = 9$ give $x = \pm 4$ and $x = \pm 3$.
$(4,\ 3)$, $(-4,\ 3)$, $(3,\ -4)$ and **$(-3,\ -4)$**

Try it: Where does $x^2 + y^2 = 20$ meet $xy = 8$?

check: **$(2,\ 4)$, $(4,\ 2)$, $(-2,\ -4)$** and **$(-4,\ -2)$**